

Malcolm Cephas

Electronics & Communication Engineer | Embedded & IoT Security

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SUMMARY

Electronics and Communication Engineering graduate with hands-on experience in IoT security, embedded systems, cryptography, and software development. Experienced in designing secure access-control systems for distributed IoT environments using ECC, Attribute-Based Encryption, C++, Java, and Python. Familiar with I2C, UART, wireless networking, and secure communication architectures.

EDUCATION

Bachelor of Engineering – Electronics and Communication Engineering

2022 - 2026

Dayananda Sagar College of Engineering, Bangalore · 8.2 CGPA

EXPERIENCE & INTERNSHIPS

AI Intern

Jan 2026 - Present

Sensei Technologies · Onsite, Bangalore

- Developed backend services using Java, Spring Boot, Spring AI, and Model Context Protocol (MCP).
- Integrated AI systems with Claude Desktop, Groq, OpenRouter, and Ollama for conversational interaction and tool orchestration.
- Developed an AI chatbot with persistent chat memory, multi-chat functionality, and conversation summarization.
- Built metadata-driven workflows and schema-aware SQL generation using structured tool definitions and database metadata.

Java Development Intern

Oct 2025 – Dec 2025

Elevate Labs · Remote, Bangalore

- Developed six Java applications covering CRUD operations, file handling, object-oriented programming, data structures, and GUI development.
- Built command-line management applications and a Java Swing To-Do List with task creation, deletion, persistence, and automatic loading of saved tasks.

PROJECTS

Secure Access Control Framework for Medical IoT Systems

React · Spring Boot · Cryptography · C++ · Java · Python

- Designed an ECC-based access-control system for distributed IoT networks, securing data access across resource-constrained devices.
- Implemented authentication and access-control logic using C++, Java, and Python, with I2C and UART communication between embedded nodes.
- Applied Attribute-Based Encryption (ABE) for controlled access to protected resources; awarded “Best Project of Social Relevance” at DSCE Open Day 2026.

Hybrid Image Encryption (ECC-DH)

Python · ECC-DH · Chaotic Maps · Medical Imaging · Cryptography

- Developed a hybrid medical-image encryption system combining Elliptic Curve Diffie-Hellman (ECDH) for secure key exchange with 2-D chaotic-map-based image scrambling.
- Implemented encryption and decryption workflows for medical images and evaluated security using entropy, NPCR, and UACI metrics.

Expense Splitter

Spring Boot · JavaFX · Java · Greedy Algorithms · REST APIs

- Developed a full-stack expense management application using a Java Spring Boot server and JavaFX desktop client, with a greedy algorithm to minimize settlement transactions.
- Integrated a live multi-currency exchange-rate API with offline JSON-based caching.

E-Commerce System with MCP Server Integration

React · Java · Spring Boot · MySQL · Hibernate · Spring AI · MCP

- Developed a full-stack e-commerce application with a ReactJS frontend and Java Spring Boot backend, including REST APIs for application logic.
- Implemented MySQL persistence using Hibernate and integrated Spring AI with LLM services for AI-powered functionality.
- Integrated an MCP server with custom tools and prompts to enable conversational interaction with the e-commerce system.

SKILLS

C++	Python	Java	JavaScript	SQL	Embedded Software	IoT	I2C	UART	Wireless Networking
System Monitoring	Test Engineering	Cryptography	Network Security	ECC	Attribute-Based Encryption				
PKI	Encryption	Data Security	MATLAB	Simulink	Signal Processing	Circuit Analysis			
Telecom Engineering	Power Electronics	Git	GitHub	Docker	Spring Boot	React.js	MySQL		
PostgreSQL	Postman								

CERTIFICATIONS

- An Introduction to Cryptography — University of Leeds / Coursera
- IoT Devices — University of Illinois / Coursera
- MATLAB Onramp — MathWorks
- Simulink Onramp — MathWorks

LANGUAGES

English, Kannada, Tamil